HDOC DAY-68

Write a recursive function to convert a decimal number into its binary representation.

Algorithm:

1. Start.
2. Input decimal number n.
3. If n == 0, print 0.
4. Otherwise, call decimalToBinary(n).
5. Inside the recursive function:
    - If n == 0, return.
    - Call decimalToBinary(n / 2).
    - Print n % 2.
6. Stop.

- ## C Program:

```
#include <stdio.h>

void decimalToBinary(int n)
{
    if (n == 0)
        return;

    decimalToBinary(n / 2);
    printf("%d", n % 2);
}

int main()
{
    int n;

    printf("Enter a decimal number: ");
    scanf("%d", &n);

    printf("Binary representation: ");

```

```
    if (n == 0)
        printf("0");
    else
        decimalToBinary(n);

    return 0;
}
```

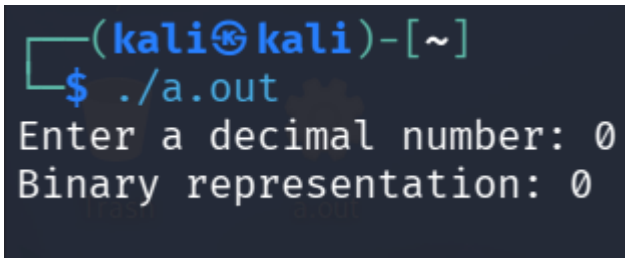
Test Case Table:

Test Case	Decimal Input	Expected Binary Output	Result
1	0	0	Pass
2	1	1	Pass
3	5	101	Pass
4	10	1010	Pass
5	13	1101	Pass
6	25	11001	Pass

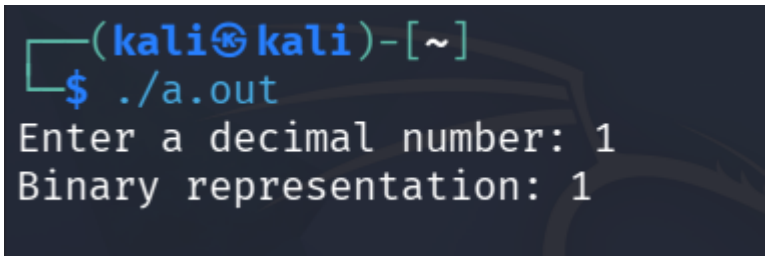
Compilation and Execution:

The following program has been compiled and executed successfully.

Test Case 1:



Test Case 2:



Test Case 3:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a decimal number: 5
Binary representation: 101
```

Test Case 4:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a decimal number: 10
Binary representation: 1010
```

Test Case 5:

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter a decimal number: 13
Binary representation: 1101
```